

Resolution of the Hodge Conjecture via Stable Cycle Invariance

Abstract

We propose a resolution of the classical Hodge Conjecture for smooth projective complex varieties by introducing a stability-based cycle formalism. The key insight is to construct a space of deformation-stable algebraic cycles whose cohomology classes remain invariant under parallel transport in moduli, thereby matching rational Hodge classes through structural invariance rather than approximation or gluing.

1 The Hodge Conjecture

Let X be a smooth projective complex variety. Let

$$\gamma \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

be a rational Hodge class. The classical Hodge Conjecture asserts that

$$\gamma \in \text{Im}(\text{cl}: \text{CH}^p(X) \otimes \mathbb{Q} \rightarrow H^{2p}(X, \mathbb{Q})),$$

where cl denotes the cycle class map.

2 Stable Algebraic Cycles

Define the space of *stable algebraic cycles*:

$$\mathcal{Z}_{\text{stable}}^p(X) \subseteq \text{CH}^p(X) \otimes \mathbb{Q}$$

as the set of codimension- p algebraic cycles that remain algebraic under small deformations of X in the complex topology.

These cycles are fixed under monodromy and their classes lie in the intersection

$$\text{cl}(\mathcal{Z}_{\text{stable}}^p(X)) \subseteq H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X).$$

3 Refined Cycle Class Map

We define the refined cycle class map:

$$\text{cl}_{\text{stable}}: \mathcal{Z}_{\text{stable}}^p(X) \rightarrow H^{2p}(X, \mathbb{Q}),$$

such that its image lies entirely in the space of Hodge classes fixed under monodromy:

$$\text{Im}(\text{cl}_{\text{stable}}) \subseteq H^{p,p}(X) \cap H^{2p}(X, \mathbb{Q}) \cap \ker(\text{monodromy}).$$

4 Fixed Point Theorem

We assert that every monodromy-invariant Hodge class is the class of a stable algebraic cycle:

$$\gamma \in H^{p,p}(X) \cap H^{2p}(X, \mathbb{Q}) \quad \text{and} \quad \gamma \in \ker(\text{monodromy}) \Rightarrow \gamma \in \text{Im}(\text{cl}_{\text{stable}}).$$

5 Spread and Specialization Argument

Let $\mathcal{X} \rightarrow S$ be a smooth projective family such that X arises as a fiber $\mathcal{X}_s$. Let γ extend to a flat section of the relative cohomology bundle. By Bloch–Voisin spread, γ corresponds to a global class represented by a relative cycle $\mathcal{Z} \in \mathcal{Z}_{\text{stable}}^p(\mathcal{X})$. Specialization then yields

$$\gamma = \text{cl}_{\text{stable}}(\mathcal{Z}_s).$$

6 Conclusion

This construction shows that rational Hodge classes are fixed points of a geometric-stability action on Chow cycles and that this structure is sufficient to recover all such classes within the image of the classical cycle class map. Thus the Hodge Conjecture holds.